TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Audio-Augmented Digital Twin Framework for Smart Manufacturing Systems			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical · Electronics		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Measuring Instruments	Safety Monitoring / Diagnosis Control Instruments	R&D / Engineering Services	Testing / Inspection / Analysis Techniques
Concept of Proposed Technology	Current digital twin systems are widely used in smart manufacturing to mirror physical assets and processes in virtual environments. However, most existing digital twins rely primarily on structured data sources, such as control signals, numerical sensor readings, and predefined simulation models. As a result, they often lack the capability to adequately reflect subtle state changes, environmental influences, and real-time interactions occurring in complex industrial systems. Recent studies on digital twins in manufacturing have similarly noted that conventional digital twin architectures remain largely dependent on structured sensor data and predefined models, limiting their ability to capture complex and context-dependent system behaviors [1,2]. This project proposes a general-purpose digital twin framework that extends conventional digital twins with an audio-based perception and interaction layer. Sound is introduced as a complementary and non-invasive modality that captures rich information about system behavior,			

	operational dynamics, and environmental context. Rather than targeting specific detection or diagnosis tasks, the proposed framework focuses on enabling digital twins to more coherently perceive, reflect, and interact with physical systems across diverse operating conditions. By augmenting digital twins with audio-based perceptual cues, the framework supports improved state awareness, consistency between physical and digital representations, and more intuitive interaction with human operators and higher-level control systems. Audio signals provide a continuous and system-level view of operational conditions, making them particularly suitable for enhancing digital twins in safety-critical and autonomous manufacturing environments. This perspective is aligned with recent research that explores perception-oriented and cognitive digital twin concepts, where additional sensing modalities are incorporated to improve state awareness and interaction capabilities [3,5]. The proposed technology is not tied to a specific machine, process, or industrial sector. Instead, it establishes general principles and software components for integrating audio-based perception into digital twin architectures, enabling reuse across multiple manufacturing scenarios. This approach aligns with the goals of smart safety and intelligent manufacturing by facilitating more responsive, interpretable, and human-centric digital twin systems.
Rationale for Joint R&D Project	Digital twin technologies for smart manufacturing are inherently system-level and context-dependent, as they aim to reflect and interact with complex physical environments in real time. Developing a general-purpose digital twin framework that goes beyond isolated simulations requires validation across diverse industrial settings, operational practices, and sensing infrastructures. Such diversity cannot be sufficiently addressed within a single institution or a single national context. Prior digital twin studies have emphasized that system behavior, sensing characteristics, and deployment constraints can vary significantly across industrial contexts, underscoring the need for evaluation beyond isolated or single-site settings [1,4]. In particular, the integration of audio as a perceptual and interaction modality for digital twins introduces challenges related to environmental variability, cultural differences in industrial practices, and heterogeneous deployment conditions. Acoustic characteristics are strongly influenced by site-specific factors, including machine layouts, building structures, and operational routines. International joint research is therefore essential to ensure that the proposed audio-augmented digital twin framework remains robust, transferable, and not tailored to a specific site or vendor environment. Through international joint R&D, this project enables systematic investigation of how audio-based perception can be incorporated into digital twin architectures under heterogeneous manufacturing conditions. The collaboration allows complementary strengths to be combined: research institutes contribute foundational framework design, abstraction principles, and evaluation methodologies, while industrial partners provide practical constraints, real-world operational insights, and validation environments.

	This joint approach reduces technical and operational uncertainty by exposing the proposed framework to a wide range of manufacturing contexts from an early stage. As a result, the project is positioned to deliver a reusable and scalable digital twin framework that supports smart manufacturing and safety objectives across different industries, rather than a narrowly optimized solution for a single application scenario.
Key Words	1. Digital Twin Framework 2. Smart Manufacturing Systems 3. Audio-Based Perception 4. Human-Centric Industrial Intelligence 5. Smart Safety

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 3	End of the Project	Level 5
R&D Objectives	The overall objective of this project is to establish a reusable digital twin framework augmented with audio-based perception and interaction capabilities for smart manufacturing systems. The project focuses on defining architectural principles and software components that enable digital twins to more coherently perceive physical systems and interact with their environments, rather than addressing isolated applications or task-specific solutions. Similar directions have been discussed in recent digital twin research, which highlights the importance of perception, cognition, and human-centered interaction as key elements of next-generation digital twin architectures [3–5]. To achieve this objective, the project pursues the following specific goals: 1. Establish architectural principles for audio-augmented digital twins: Define system-level design principles that incorporate audio as a complementary perceptual modality within digital twin architectures, clarifying its role in state representation, coherence, and interaction across diverse manufacturing configurations. 2. Develop reusable software components for audio-based perception integration: Design and implement modular software components that enable the integration of audio signals as a generic perception layer within digital twin systems, supporting reuse across different industrial environments and deployment conditions. 3. Enhance state coherence between physical systems and digital twins: Investigate how audio-based perceptual cues can support consistency between physical system behavior and digital twin representations under varying operational conditions and system dynamics. 4. Support interpretable interaction with human operators and supervisory systems: Explore interaction mechanisms through which audio-augmented digital twins can provide intuitive and interpretable feedback to human operators and higher-level control systems, contributing to smart safety and human-centric manufacturing objectives. 5. Validate framework applicability in representative manufacturing environments: Demonstrate the feasibility and reusability of the proposed framework through validation in representative manufacturing scenarios, reflecting realistic system behaviors.			
R&D Target Specifications	The target performance specifications of this project are as follows: 1. Framework Integration Successful integration of audio-based perception modules within a digital twin framework, supporting at least 3 standardized interfaces with a module dependency ratio of less than 30%. 2. State Coherence Improvement Demonstration of at least 30% reduction in state divergence between physical system observations and digital twin representations under representative operating conditions. 3. Reusability Across Scenarios Application of the framework to at least two representative manufacturing scenarios, requiring less than 20% code modification per scenario. 4. Interpretability and Interaction Achievement of an operator agreement rate of 70% or higher in interpreting audio-related digital twin state indicators, with feedback latency below 1 second.			

R&D Plan	**Phase 1: Preliminary Study and Conceptual Design (Months 1–6)** Objective: Establish the conceptual foundation and feasibility of an audio-augmented digital twin framework. Key Activities: • Review and consolidation of existing digital twin architectures and perception-driven digital twin concepts • Definition of the role of audio as a perceptual and interaction modality within digital twin systems • Design of a high-level framework architecture, including perception, state representation, and interaction layers • Identification of key technical requirements and evaluation criteria for subsequent development phases Outcome: • Conceptual architecture of the audio-augmented digital twin framework • Initial feasibility assessment and baseline specifications for Phase 2 development **Phase 2: Framework Development and Validation** Year 1: Core Framework Development Objective: Develop the core software framework for integrating audio-based perception into digital twin systems. Key Activities: • Implementation of modular software components for audio-based perception integration • Definition and implementation of standardized interfaces between digital twin core functions and audio modules • Initial validation in laboratory environments using representative system configurations Outcome: • Prototype-level digital twin framework with audio-augmented perception capabilities • Verification of modularity and integration feasibility (TRL 4) Year 2: Interaction and State Coherence Enhancement Objective: Enhance state coherence and interaction capabilities of the audio-augmented digital twin framework. Key Activities: • Refinement of mechanisms for maintaining consistency between physical system behavior and digital twin representations • Development of interpretable interaction and feedback mechanisms for human operators and supervisory systems • Evaluation of framework behavior under varying operating conditions and system dynamics Outcome: • Improved framework robustness and interpretability • Demonstrated contribution of audio-based perception to state coherence and interaction quality Year 3: Validation in Representative Manufacturing Environments Objective: Validate the applicability and reusability of the proposed framework in representative manufacturing environments. Key Activities: • Adaptation and deployment of the framework in collaboration with industrial partners • Validation across multiple representative manufacturing scenarios with different system configurations • Quantitative assessment against target specifications and TRL advancement criteria Outcome:

	• Validation of the audio-augmented digital twin framework in relevant environments (TRL 5) • Finalized framework documentation and guidelines for reuse and extension

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will play a central role in establishing the scientific and technical foundation of the proposed audio-augmented digital twin framework. Its primary responsibility is to define the overall system architecture and ensure that the framework remains general-purpose, reusable, and independent of specific industrial applications or vendor platforms. In particular, the research institute will lead the conceptual design of the audio-based perception and interaction layers within the digital twin, leveraging its expertise in industrial acoustics and audio intelligence. This includes defining abstraction principles, modular interfaces, and evaluation methodologies that enable audio to function as a complementary perceptual modality in digital twin systems. The institute will also be responsible for developing prototype implementations, conducting laboratory-level validation, and coordinating framework-level evaluations across representative scenarios. Through these activities, the research institute ensures the technical coherence, scientific rigor, and long-term extensibility of the proposed framework, while enabling effective collaboration with industrial partners during later validation stages.
Expected R&R of Korean SME Partner	The Korean SME partner is expected to play a key role in validating the proposed audio-augmented digital twin framework in relevant manufacturing environments and ensuring its practical applicability. While the framework itself is designed to be general-purpose, industrial validation requires realistic system configurations, operational constraints, and deployment conditions that can only be provided by industrial partners. In particular, the SME partner will contribute domain knowledge related to manufacturing operations, system integration, and practical deployment considerations. This includes providing representative system scenarios, data interfaces, and feedback on usability, interpretability, and integration requirements from an industrial perspective. The SME partner will also participate in pilot-level validation activities during the later stages of the project, supporting the evaluation of the framework in relevant environments and contributing to the assessment of its reusability across different system configurations. Through this role, the SME partner ensures that the proposed framework is not only technically sound but also aligned with real-world manufacturing practices and constraints.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Outcomes** The project is expected to deliver a validated audio-augmented digital twin framework that extends conventional digital twin architectures with perception and interaction capabilities based on industrial audio. The primary outcome will be a reusable framework design, including modular software components, standardized interfaces, and architectural guidelines that enable the integration of audio-based perception into digital twin systems. In addition, the project will produce prototype implementations and validation results demonstrating the applicability of the framework in representative manufacturing environments. These outcomes will be accompanied by technical documentation and evaluation results that support reuse and

	further extension by industrial partners and system integrators. **Expected Impact** By introducing audio as a complementary perceptual modality, the proposed framework expands the functional scope of digital twins beyond simulation and monitoring toward more responsive, interpretable, and human-centric smart manufacturing systems. The project contributes to improving state awareness and interaction capabilities in digital twins, which are critical for smart safety and autonomous manufacturing environments. From an industrial perspective, the outcomes provide a foundation for developing more adaptable and cost-efficient digital twin solutions that can be applied across diverse manufacturing contexts. From a research and policy perspective, the project strengthens international collaboration in digital twin technologies and supports Korea's strategic goals in smart manufacturing, industrial intelligence, and safety-oriented digital transformation.

6. References

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(3) Luo, Junjie, et al. "A perception-powered urban digital twin to support human-centered urban planning and sustainable city development." Cities 156 (2025): 105473.

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(5) Al Faruque, Mohammad Abdullah, et al. "Cognitive digital twin for manufacturing systems." 2021 Design, Automation & Test in Europe Conference & Exhibition (DATE). IEEE, 2021.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization